Anshul Raj Singh

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Education

Bachelor of Mathematics (Honours) *January 2026 - Present*

Indian Statistical Institute, Bangalore

Expected graduation: *May 2027*
Current cumulative grade: 87%

Independent study and reading projects

- **Reading Project in Analytic Number Theory** *June 2026*
  Studied the Distribution of primes, and Hardy–Littlewood's circle method, and wrote an exposition on the proof of Vinogradov's Three-Primes theorem under the guidance of Dr. Alessandro Fazzari, University of Genoa, Italy.
- **Reading Project in Representation Theory of Finite Groups** *December 2025*
  Studied J. P. Serre's *Linear Representations of Finite Groups* under the guidance of Prof. Manish Kumar at Indian Statistical Institute, Bangalore.

Writings & Preprints

- **Vinogradov's three prime theorem** – [PDF] *June 2026*
  We explore the distribution of prime numbers, proving the Prime Number Theorem, and the Prime Number Theorem for arithmetic progressions. We then introduce the Hardy–Littlewood circle method and present a detailed proof of the Vinogradov's Three-Primes theorem thereby establishing the ternary Goldbach problem.
- **On a square packing conjecture of Erdős** – [arXiv:2601.22163] *January 2026*
  Established an equivalent formulation of an Erdős square-packing conjecture Problem 106 and investigate related extremal packing problems using an inequality due to Praton.

Talks Presented

- **Construction Geometry – Ruler and Compass constructions** *August 2026*
  Presentation on ruler and compass constructions and the Poncelet–Steiner theorem as part of the Galois Theory course at the Indian Statistical Institute, Bangalore.
- **Circle Method and the Vinogradov's Three-Prime Theorem** *October 2026*
  A talk on the Hardy–Littlewood circle method and the sketch of the proof of Vinogradov's Three-Prime presented to ISI Bangalore Math Club.

Programming and Mathematical tools

- **Programming:** Python, R.

- **Mathematical software:** NumPy, Jupyter Notebook, LaTeX.